Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

AI

Practical IV Task

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Class: TYIT	Batch:1
Date of Assignment: 25-07-2026	Date/Time of Submission: 25-07-2026

Q.1 In a college:

- 20% of students know Python.
- 80% of students know Java.
- 90% of Python students get placed.
- 40% of Java students get placed.

A randomly selected student is placed.
Find:

- Probability that the student knows Python.
- Probability that the student knows Java.
- Probability that a placed student knows Python.

Code:-

```python
python = 20

java = 80

python_placed = 90

java_placed = 40

p_python = python / 100

p_java = java / 100

placed_python = p_python * (python_placed / 100)

placed_java = p_java * (java_placed / 100)

total_placed = placed_python + placed_java

p_python_given_placed = placed_python / total_placed

print("Probability of Python =", p_python)

print("Probability of Java =", p_java)

print("Probability that a placed student knows Python =", p_python_given_placed)
```

Output:-

```
Probability of Python = 0.2
Probability of Java = 0.8
Probability that a placed student knows Python = 0.36
```

Q.2 In a city:

- 85% of taxis are Green.
- 15% are Blue.
- A witness correctly identifies the taxi color 80% of the time.

A witness claims the taxi involved in an accident was Blue. What is the probability that the taxi was actually Blue?

Code:-

```python
green = 0.85

blue = 0.15

correct = 0.80

wrong = 0.20

# Probability that taxi is actually Blue

p_blue = (blue * correct) / ((blue * correct) + (green * wrong))

print("Probability that the taxi was actually Blue =", round(p_blue, 3))
```

Output:-

```
Probability that the taxi was actually Blue = 0.414
```